SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	IV - Curves, Tacheometry & Modern Methods

I. Curves - Types and Necessity

A **curve** is a smooth transition between two straight lines (tangents) of a road, railway or canal alignment that change direction. Curves provide a gradual change of direction for safety, comfort and economy.

- Necessity: to change the direction of the alignment smoothly; to avoid sudden jerks; to allow safe design speed; to adapt to the topography.

Classification

Type	Description
Horizontal curves	Change of direction in the horizontal plane: simple, compound, reverse, transition (spiral), lemniscate
Vertical curves	Change of gradient in the vertical plane: summit (crest) and sag (valley) curves

- **Simple curve:** a single circular arc connecting two tangents.
- **Compound curve:** two or more simple arcs of different radii, curving the same way and meeting at a common tangent point (P.C.C.).
- **Reverse curve:** two arcs of the same or different radii, curving in opposite directions (S-shape); used in hilly roads and in crossovers on railways.
- **Transition curve:** a curve of varying radius (cubic parabola / spiral / lemniscate) inserted between a straight and a circular curve - gives gradual superelevation and centrifugal force. Not in the syllabus in detail.

2. Elements of a Simple Circular Curve

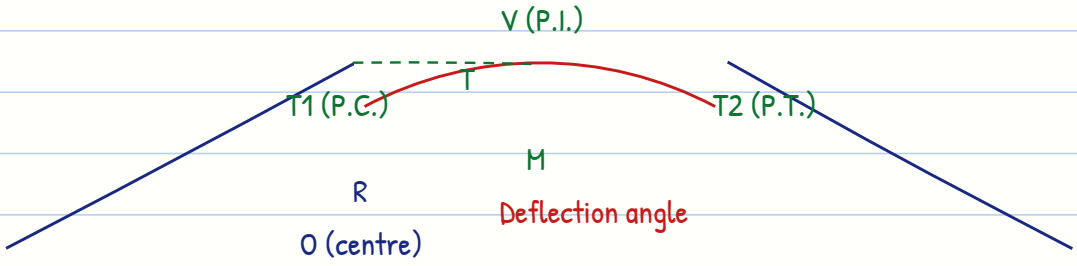


Fig 1: simple circular curve

Element	Meaning / formula
Tangents	Back tangent (AB) and forward tangent (BC) meeting at the point of intersection V (P.I.)
Deflection angle (D)	Angle by which the forward tangent deviates from the back tangent produced
Angle of intersection	180 deg - D
Point of curve (T1)	Beginning of the curve (P.C.)
Point of tangency (T2)	End of the curve (P.T.)
Tangent length (T)	T = R tan(D/2)
Length of curve (l)	l = pi R D / 180 = R x D (radians)
Long chord (L)	L = 2 R sin(D/2)
Apex distance	Ext. = R [sec(D/2) - 1]
Versed sine (M)	M = R [1 - cos(D/2)]
Degree of curve (Dc)	Angle subtended at the centre by a 30 m arc: R = 1718.9 / Dc

R = 1718.9 / Dc (metres, 30 m arc basis)

Chainage of T1 = Chainage of V - T

Chainage of T2 = Chainage of T1 + l

KEY:
Deflection angle at the apex = central angle of the curve. Tangents are equal: VT1 = VT2 = R tan(D/2). The length of the curve is the arc length T1 to T2.

Designation of curve

- **By radius:** the curve is specified by its radius, e.g. 300 m curve (used in roads).
- **By degree of curve:** angle subtended by a standard chord (30 m in metric railway practice, 20 m in India for railways).
- Sharpness of a curve: the smaller the radius (larger degree), the sharper the curve.

3. Setting Out a Simple Curve

Linear (chain and tape) methods:

- **Offsets from the long chord:** ordinates measured from the long chord at regular intervals; O_x = sqrt(R^2 - x^2) - (R - M), x measured from the mid-point of the chord; the central ordinate = M (versed sine).
- **Successive bisection of the arcs:** mid-point offsets are found by successive halving of the chord (versed sine).
- **Offsets from the tangents:** perpendicular offset O_x = R - sqrt(R^2 - x^2) (exact) or x^2 / 2R (approx.); radial offset O_x = sqrt(R^2 + x^2) - R.
- **Offsets from chords produced:** each peg is set with an offset from the chord produced; O1 = C1^2 / 2R, O2 = C2 (C1 + C2) / 2R.

Angular (instrumental) methods:

- **Rankine's method of deflection angles:** a theodolite at T1 sets pegs by the deflection (tangential) angle from the tangent; the most common method.
- **Two theodolite method:** two theodolites at T1 and T2 intersect at each peg; used where chaining is difficult (water, obstacles).
- **Tacheometric method and the total station method:** with polar coordinates.

Rankine's deflection angle method

The deflection (tangential) angle for a chord of length c is the angle between the tangent and the chord. It is half the central angle of the chord.

delta = (c / 2R) radians = 1718.9 c / R minutes

Total deflection angle at T2 = D / 2

First chord c1 (sub-chord), normal chords c = 30 m, last chord c2

- Deflection angle for the first sub-chord: delta1 = 1718.9 c1 / R (min); each normal chord: delta = 1718.9 c / R (min).
- Total deflection at the end of the curve = sum of the deflection angles = D/2. This is a ***check*** on the work.
- Procedure: set the theodolite at T1; with A at zero, sight V (tangent direction); turn by delta1; measure c1 from T1 to fix point 1; then turn by delta1 + delta2; swing the chain from peg 1 with length c; continue until T2.

Example

Two straights meet at chainage 1190 m with a deflection angle of 36 deg. Radius = 300 m. Find the tangent length, length of curve, and chainages of T1 and T2.

T = 300 tan 18 deg = 97.48 m

l = pi x 300 x 36 / 180 = 188.50 m

Chainage T1 = 1190 - 97.48 = 1092.52 m

Chainage T2 = 1092.52 + 188.50 = 1281.02 m

Obstacles in setting out curves

- When the point of intersection is inaccessible (e.g. in a lake or a building), the tangent lines are joined by a line connecting two points on tangents; the angles are measured and the deflection angle and the tangent length found by the sine rule.
- When the curve cannot be set out from one point because of an obstruction, the curve is set out from two or more stations (stage setting).

4. Compound Curve

A compound curve has ***two circular arcs*** with radii R1 and R2 in the same direction, with a common tangent at the point of compound curvature (P.C.C.). Elements: R1, R2, D1, D2, D (total deflection angle), the tangents VT1, VT2 and the common tangent t.

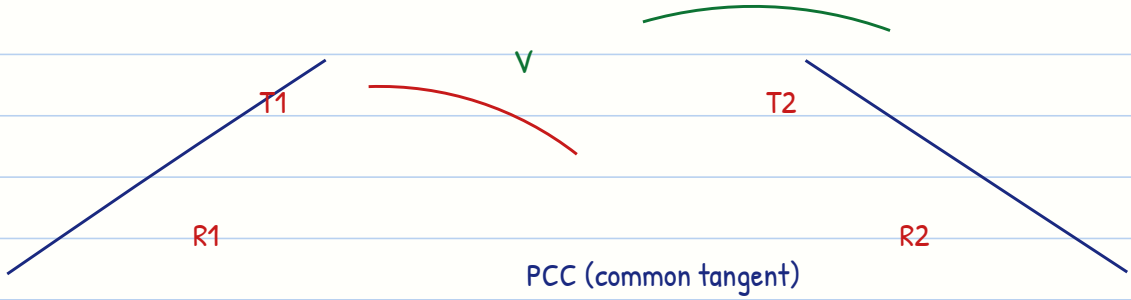


Fig 2: compound curve

$$D = D1 + D2$$

$$Ta = R1 \tan(D1/2);\ Tb = R2 \tan(D2/2);\ \text{common tangent } t = Ta + Tb$$

$$VT1 = Ta + t \sin(D2)/\sin(D);\quad VT2 = Tb + t \sin(D1)/\sin(D)$$

- Given D, R1, R2 and D1 (or D2) -> compute the rest. Used where the topography prevents a single curve; ***flat arc at the ends, sharper in the middle*** is preferred.
- Setting out: Rankine's method from T1 up to the point of compound curvature and then from T2 (back) or from the P.C.C. after orienting.

5. Reverse Curve

A reverse curve consists of two arcs of ***opposite curvature*** , joined at the ***point of reverse curvature (P.R.C.)*** where they have a common tangent. Used for connecting two parallel (or non-parallel) straights.

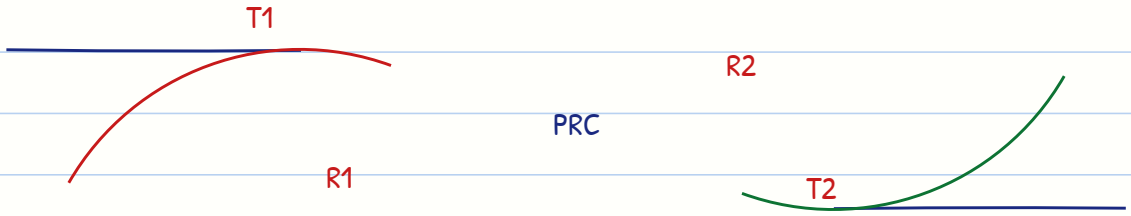


Fig 3: reverse curve between parallel straights

Two parallel straights a distance d apart, each arc subtending angle D:

$$d = (R1 + R2)(1 - \cos D)$$

If R1 = R2 = R: $d = 2R(1 - \cos D)$; length of curve = $2 \times \pi R D / 180$

- Common tangent at the PRC; the centres of the two arcs lie on the opposite sides of the common tangent.

- Reverse curves are ***not used*** in highways and fast railways because of the sudden change in the sign of superelevation (and the passengers feel discomfort); used for slow-moving roads, hills, yards.

> **COMPARE:**
> Compound curve: same side centres. Reverse curve: centres on opposite sides.
> Simple: one arc. Transition: varying radius.

6. Introduction to Tacheometric Surveying

Tacheometry is a rapid method of determining ***horizontal distances and elevations*** by optical means, using a ***tacheometer*** (a theodolite fitted with a stadia diaphragm) and a graduated staff. No chaining or levelling is needed.

- Used for: topographical surveys in hilly, difficult terrain; rough locations; checking chain surveys; contouring; traverse in rugged areas.
- Accuracy about 1 in 500 to 1 in 1000, lower than chaining.

Stadia method: principle

The ***stadia hairs*** (upper and lower) in the diaphragm intercept a length (staff intercept s) on a levelling staff at a distance D. The ratio D / s is constant for a given instrument.

Horizontal sight (staff vertical):

$$D = (f/i)\, s + (f + d) = K s + C$$

K = f/i = multiplying constant (usually 100), C = additive constant (usually 0 for an anallactic telescope)

- ***Anallactic lens:*** extra lens in the telescope that makes C = 0, so D = K s, simplifies the work.
- Here f = focal length of the objective, i = stadia hairs interval, d = distance from the objective to the vertical axis.

Inclined sight, staff vertical

Horizontal distance D = K s cos^2(theta) + C cos(theta)

Vertical distance V = (K s / 2) sin(2 theta) + C sin(theta)

R.L. of staff station = R.L. of instrument axis + V - h (for elevation)

(theta = angle of elevation; h = axial staff reading, i.e. middle hair)

- For angle of depression, V is negative: R.L. = R.L. (axis) - V - h.
- With C = 0: D = K s cos^2 theta; V = (K s / 2) sin 2 theta.

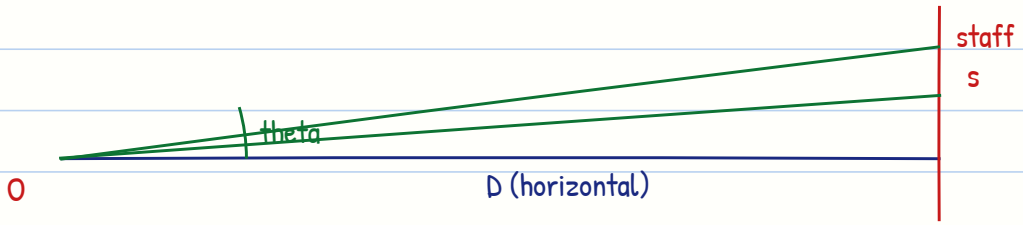


Fig 4: stadia tacheometry, inclined sight

Determination of the constants K and C

- Place the staff at two known distances D1 and D2 (e.g. 100 m and 200 m) from the instrument; read the intercepts s1 and s2.

K = (D2 - D1) / (s2 - s1)

C = D1 - K s1

Other systems of tacheometry

- ***Tangential method:*** staff with two targets at a known distance s apart; vertical angles to the targets give D = s / (tan alpha1 - tan alpha2).
- ***Subtense bar method:*** a horizontal bar of known length (2 m) at the station; the small horizontal angle subtended at the instrument gives distance D = (b/2) cot (beta/2).
- ***Moving hair method:*** stadia hairs can be moved; the movement is read from the screw head.

Example

A tacheometer with K = 100 and C = 0 is set at a station. At the staff reading with a vertical angle of +5 deg: upper hair 2.355, middle 1.905, lower 1.455. Staff intercept s

= 0.900 m.

D = 100 x 0.9 x cos^2 5 deg = 89.32 m
V = 100 x 0.9 / 2 x sin 10 deg = 7.81 m

7. Modern Surveying – EDM (Electromagnetic Distance Measurem

An **EDM** instrument measures the distance by the ***time of travel / phase difference*** of electromagnetic waves between the instrument and a reflector, in seconds. Quick, accurate (1 mm +/- 1 ppm), long range.

Principle

Distance D = (c x t) / 2 (pulse method)
c = velocity of light in the medium (approximately 3 x 10^8 m/s)

- Wave sent from the transmitter reflects at the target prism / reflector and returns; the ***time or phase difference*** is measured automatically and converted to distance.

Types of EDM instruments

Type	Wave / range
Microwave instruments (Tellurometer)	Microwaves; range up to 100 km; two instruments (master and remote); affected by weather less
Infrared (EDM) instruments	Infrared waves modulated; reflector prism; range 2-5 km; widely used with total station
Light wave (laser) instruments (Geodimeter)	Visible laser / light; range 2-3 km by prism; also reflectorless to 200-1000 m

- ***Phase comparison method:*** the phase difference between the sent and the received wave gives the fractional wavelength; several modulation frequencies resolve the integer ambiguity.
- ***Pulse method:*** the time of flight of short pulses is directly measured (used in reflectorless EDM).
- Corrections: atmospheric (temperature, pressure, humidity), instrument constant, prism constant, slope reduction, curvature, sea-level reduction.

8. Total Station

A ***total station*** combines an ***electronic theodolite*** , an ***EDM*** and a ***microprocessor*** with data storage in one instrument – it measures angles and distances and computes coordinates automatically.

- Parts: telescope with EDM, electronic angle encoder, microprocessor, keypad, display, data collector (memory card), battery, tribrach, prism reflector with pole.
- Measures horizontal angle, vertical angle, slope distance -> computes horizontal distance, vertical distance (height), and the ***northing, easting, elevation*** of the observed point.

Advantages

- Fast, accurate, and automatic data recording – minimum human error; transfer to a computer / CAD directly.
- Reflectorless measurement; long range; one instrument for many tasks; can do setting out; resection and free-station measurements.
- Reduced fieldwork and manpower; automatic correction for the curvature, refraction, temperature and pressure.

Applications

- Topographic and contour surveys; traversing and control; setting out of buildings, roads, bridges, tunnels; deformation monitoring of dams, buildings.
- Volume calculation (stockpile / earthwork), as-built surveys, mine surveys, cadastral and boundary surveys.

Field procedure (brief)

- Set up and level over the station; enter the instrument height, prism height, temperature, pressure; orient to a backsight; measure to the points; store and download.

9. Global Positioning System (GPS)

GPS is a ***satellite-based*** navigation and positioning system, operated by the USA, that gives the 3-D position (latitude, longitude, height) and time anywhere on or near the earth. ***GNSS*** is the general term for all such systems: GPS (USA), GLONASS (Russia), Galileo (Europe), BeiDou (China), ***NavIC (India)*** .

Three segments

- **Space segment:** 24+ satellites in 6 orbital planes at about 20,200 km altitude, period 12 hours; at least 4 satellites visible anywhere.
- **Control segment:** master control station and monitor stations that track the satellites, compute orbits and upload the data.
- **User segment:** receivers (handheld, geodetic) used by surveyors, navigators.

Working principle (trilateration)

Each satellite broadcasts its position and the time of transmission. The receiver measures the signal travel time and gets the distance (pseudo-range) = c x time. Distance from **at least 4 satellites** fixes x, y, z and the receiver clock error.

- **Static GPS:** two or more receivers occupy points for a long time; cm or mm accuracy. Used for control surveys.
- **Kinematic / RTK (Real Time Kinematic):** a base and a moving rover with radio link; 1-2 cm accuracy in real time.
- **DGPS:** corrections from a base station improve the accuracy to sub-meter.

Errors in GPS

- Satellite and receiver clock errors; orbit errors; ionospheric and tropospheric delay; multipath; poor satellite geometry (high DOP); selective availability (earlier).
- Applications: control surveys, geodetic surveys, cadastral and engineering surveys, navigation, deformation monitoring, GIS data collection, precision agriculture.

10. Drone (UAV) Survey

A **drone survey** uses an **Unmanned Aerial Vehicle (UAV)** carrying a camera (or LiDAR) to collect aerial images and data for mapping. Images are processed by **photogrammetry** software to give orthomosaics, point clouds, DEM, DSM and 3-D models.

- Workflow: flight planning -> ground control points (GCPs, measured by GPS) -> image capture with overlap (80 % forward and 60-70 % side) -> processing (aerial triangulation, dense cloud) -> outputs.
- Types: fixed wing (large areas, longer flights), multi-rotor (small areas, can hover) and hybrid VTOL.

- Advantages: quick, cheap, high resolution (1–5 cm per pixel), access to dangerous areas, frequent monitoring, safer for personnel.
- Limitations: battery life, weather, regulations (licenses and permits – India: Drone Rules 2021), accuracy depends on GCPs, vegetation reduces accuracy.
- Applications: topographic mapping, volume of stockpiles, road and railway corridor surveys, land records (SVAMITVA scheme), mining, construction progress monitoring, disaster assessment.

II. LiDAR Survey (Light Detection And Ranging)

LiDAR is an active remote sensing technique. A laser scanner emits rapid pulses of light (up to hundreds of thousands per second) towards the ground; the ***time of return*** of each pulse gives the distance; with the GPS position and the IMU (orientation) of the platform, each return is a 3-D point (x, y, z) – producing a ***point cloud*** .

Range R = (c x t) / 2

Point position = GNSS position + range vector rotated by IMU angles

- Platforms: ***airborne*** (aircraft / drone), ***terrestrial*** (tripod / static) and ***mobile*** (vehicle-mounted) LiDAR.
- Output: dense point cloud; ***DSM*** (digital surface model) and ***DTM / bare-earth model*** by classifying ground returns; contours, 3-D city models.
- Advantages: very high accuracy (a few cm); works through gaps in vegetation (multiple returns); day or night; fast for large areas; detailed terrain in forests and in corridors.
- Limitations: costly equipment; large data volume to process; poor in heavy rain / fog; dark or reflective surfaces.
- Applications: flood modeling, highway and railway design, power line mapping, forest inventory, coastal and urban planning, archaeology, mining volume.

DIFFERENCE:

Drone survey = photographs (passive). LiDAR = laser pulses (active). Both give DEM / point clouds; LiDAR penetrates vegetation, photogrammetry cannot.

12. Worked Example – Deflection Angle Table

Using the earlier data (R = 300 m, D = 36 deg, chainage of T1 = 1092.52 m, T2 = 1281.02 m, peg interval 30 m). The first sub-chord = 1110 – 1092.52 = 17.48 m; last sub-chord = 1281.02 – 1260 = 21.02 m.

Peg	Chainage	Chord (m)	Deflection (min)	Cumulative
T1	1092.52	–	0	0 00
1	1110.00	17.48	100.15	1 40 09
2	1140.00	30.00	171.89	4 32 02
3	1170.00	30.00	171.89	7 23 55
4	1200.00	30.00	171.89	10 15 48
5	1230.00	30.00	171.89	13 07 41
6	1260.00	30.00	171.89	15 59 34
T2	1281.02	21.02	120.44	18 00 00 (= D/2)

CHECK:

Check: the final cumulative deflection must equal D/2 = 18 deg. If not, there is an arithmetic error – recheck the sub-chords.

13. Superelevation and Transition (short note)

- **Superelevation (cant):** raising the outer edge of the road / rail on a curve to balance the centrifugal force. e = V^2 / (127 R) for roads (V in km/h, R in m); cant for railways = G V^2 / (127 R).
- **Transition curve:** needed to introduce the radius gradually, to provide the superelevation gradually, and to avoid sudden lateral jerk. Length L = V^3 / (C R) (Shortt's formula).
- **Vertical curves:** parabolic curve joining two grades; ***summit*** (crest) curve when the gradient decreases, and ***sag*** curve when it increases. Rate of change of grade is kept small for comfort and sight distance.
- Typical grade change: length of a vertical curve L = N x S, where N = algebraic difference of the gradients, S = the sight distance factor.

14. Another Tacheometry Problem

A tacheometer set at P (R.L. of axis = 450.00 m), K = 100, C = 0. Staff held vertically at Q: vertical angle = -3 deg; readings: 1.200, 2.050, 2.900. Find D and R.L. of Q.

s = 2.900 - 1.200 = 1.700 m

D = 100 x 1.7 x cos^2 3 deg = 169.53 m

V = 100 x 1.7/2 x sin 6 deg = 8.88 m (depression)

R.L. of Q = 450.00 - 8.88 - 2.050 = 439.07 m

- Remember: for depression angle, subtract V and subtract the axial reading h from the R.L. of the axis.
- When the staff is held normal to the line of sight (instead of vertical): D = K s cos(theta) + C cos(theta) + h sin(theta); V = K s sin(theta) + C sin(theta).

15. Important Questions to Prepare

- Define a simple circular curve; derive the formulae for tangent length, length of curve and long chord.
- Explain the Rankine's deflection angle method of setting out a circular curve with the help of a table.
- Differentiate between compound and reverse curves; give their uses.
- Derive the distance and elevation formulae for the stadia method of tacheometry (inclined sight, staff vertical).
- Explain the principle and types of EDM; the advantages and applications of a total station.
- Write short notes on GPS, drone survey and LiDAR survey.

16. Quick Revision

- Simple curve: T = R tan(D/2); l = pi R D/180; L = 2R sin(D/2); chainage T1 = V - T.
- Deflection angle for a chord = 1718.9 c / R minutes; the total = D/2 (check).
- Tacheometry: D = K s cos^2 theta, V = (K s/2) sin 2 theta.
- EDM: D = ct/2; total station = theodolite + EDM + microprocessor; GPS needs minimum 4 satellites.
- Drone = camera images; LiDAR = laser point cloud.